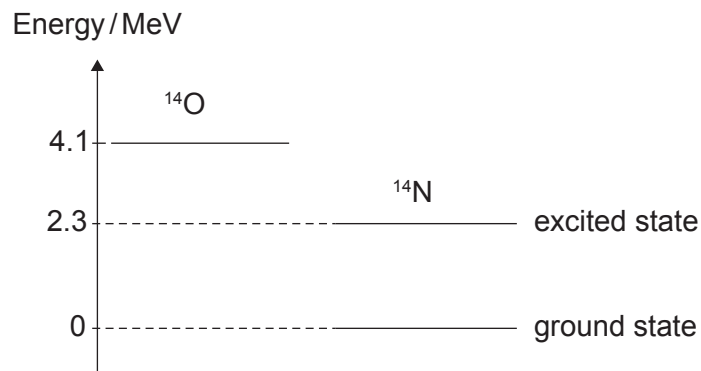


E.3 / PAPER 1 / QUESTION 32

4 November 2021 / TZ0 / HL Paper 1 / original Q39 / 8821-6501

32. Some of the nuclear energy levels of oxygen-14 (^{14}O) and nitrogen-14 (^{14}N) are shown.



A nucleus of ^{14}O decays into a nucleus of ^{14}N with the emission of a positron and a gamma ray. What is the maximum energy of the positron and the energy of the gamma ray?

	Maximum energy of the positron / MeV	Energy of the gamma ray / MeV
A.	1.8	2.3
B.	1.8	4.1
C.	2.3	1.8
D.	4.1	2.3